Find the signal. Keep the auditor in control.

FINTECH AI — Digital Surfers | Evidence-led investigation of unusual financial transactions

THE PROBLEM

- High-volume transaction spreadsheets make unusual items hard to spot.
- A raw anomaly flag lacks context and follow-up evidence.
- Review decisions and investigation history need a consistent record.

AI assists investigation; it does not decide fraud or replace professional judgment.

BEYOND BANKS: 10 BUSINESS CATEGORIES

- Retail + e-commerce: supplier / fulfilment payments
- Restaurants / hotels + logistics: operating expenses
- Schools + hospitals: procurement invoices
- NGOs + construction: project / contractor costs
- Manufacturers + audit firms: vendor / client review

Users: auditors, finance-control teams and business owners.

PROPOSED SOLUTION

Import → screen → explain → review

Deterministic data checks work without AI. Gemma is an optional evidence-grounded explanation layer.

01 IMPORT

Validate CSV rows and persist transactions.

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02 SCREEN

Flag repeated references and supplier + amount + date matches.

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03 EXPLAIN

Gemma interprets server-supplied evidence; no database/tool access.

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04 REVIEW

Auditor confirms or dismisses; decision and timestamp are stored.

EVIDENCE BOUNDARY

Observed evidence is persisted transaction / rule output. Gemma contributes interpretation and recommended checks—not new transaction facts or financial actions.

One audit API. Cloud Gemma + local Gemma.

Next.js / React → FastAPI / Pydantic → SQLite. AI adds interpretation; deterministic screening remains independent.

FRONTEND

- Dashboard + CSV upload
- Transactions + findings
- Same-origin API proxy

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API + DATABASE

- FastAPI / HTTPX / SQLite
- Transactions + findings
- Explanations + reviews

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CLOUD: LIVE

- Google AI Studio
- gemma-4-26b-a4b-it
- App Explain uses cloud

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LOCAL: LIVE

- Ollama: gemma4:latest
- 7.5B Q4_K_M / 6.6GB
- Remote /ollama API access

REPRODUCIBLE EVIDENCE

7 backend / SDK tests passed • SDK wheel built • frontend TypeScript check passed • cloud + local inference verified. Separate pipeline sample: 100 accepted, 6 duplicate rows, 34 anomalies; full API hookup pending.

CSV → duplicate evidence → Gemma explanation → review

Public app: first-reseller-flows-statutes.trycloudflare.com/dashboard

SHOW THE END-TO-END FLOW

- Upload 01_mixed_office_transactions.csv (20 valid rows).
- Open Anomalies & AI; inspect a duplicate candidate.
- Explain: Google-hosted Gemma interprets the supplied evidence.
- Save review; refresh to show persisted status.
- Local model: POST /ollama/api/generate; Swagger: /backend/docs

THREE SYNTHETIC DEMO FILES

- 01 Office: 20 valid rows; varied costs + duplicate candidates.
- 02 Retail: 20 valid rows; inventory, logistics + duplicates.
- 03 Projects: 16 valid + 4 invalid rows; demonstrate rejection reporting.
- Large payment ≠ fraud. Same-value invoices may be legitimate; auditor checks.

CSV downloads: /demo-data/ on the same host. Public smoke test passed upload → finding → saved review; local and cloud generation both succeeded.

Built to review payments. Open for others to integrate.

Impact target: reduce spreadsheet hunting and improve review traceability; time savings and accuracy are not yet measured.

DELIVERED: REUSABLE APPLICATION CODE

- MIT LICENSE • github.com/kwisdomk/FintechClankerz • main + mbr-3.
- API v0.4.0: typed OpenAPI 3.1 contract, Swagger / ReDoc and stable operation IDs.
- Python SDK: pip install ./sdk/python; JavaScript + curl integration examples.
- docs/INTEGRATION.md • docs/openapi.json • CONTRIBUTING.md • tests + synthetic data.

HONEST LIMITATIONS / WHAT COMES NEXT

- Current screening: duplicate rules. Wire full Isolation Forest pipeline into upload next.
- Pre-1.0 MVP: add authentication, rate limits, tenant isolation and robust currency/date handling.
- DigitalOcean GPU + private Ollama: researched, NOT deployed. Benchmark a 20GB GPU before committing.
- Synthetic public demo only. Cloud sends evidence to Google; model / provider terms are separate from MIT.

Cloud hosting today: Google AI Studio. Local inference today: developer machine + tunnel. DO research: digitalocean.com/products/gpu-droplets. Stop public access after the demo.